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FIT3080 – Artificial Intelligence

Machine Learning – Part 3
Chapters 12.6, 21.2.1-21.2.2, 19.7.4, 19.8.1-19.8.4, K-Means

Learning Objectives

- **Machine learning – definition**
- **Approaches to learning from data**
- **Mathematical principles**
- **Supervised machine learning**
  - Decision Trees
  - Regression
  - Linear classifiers with thresholds
  - Logistic regression
  - Artificial Neural Networks
  - Naïve Bayes
  - k Nearest Neighbour (k-NN)
- **Unsupervised learning**
  - Clustering
    - > Similarity measures
    - > The K-means algorithm

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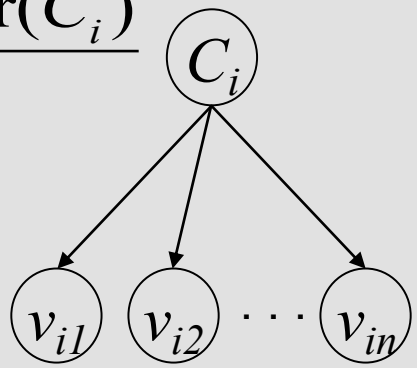
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Naïve Bayes Classifier

Naïve Bayes Classifier (I)

- **Based on Bayes rule**

$$\Pr(C_i \mid V_i) = \frac{\Pr(V_i \mid C_i)\Pr(C_i)}{\Pr(V_i)}$$



where
  - C_i is the class of item i
  - $V_i = \{v_{i1}, \ldots, v_{in}\}$ are the values of a set of attributes for item i
  - v_{ij} is the value of attribute j for item i

$$\Pr(C_i = c \mid v_{i1}, \ldots, v_{in}) = \frac{\Pr(v_{i1}, \ldots, v_{in} \mid C_i = c)\Pr(C_i = c)}{\Pr(v_{i1}, \ldots, v_{in})}$$

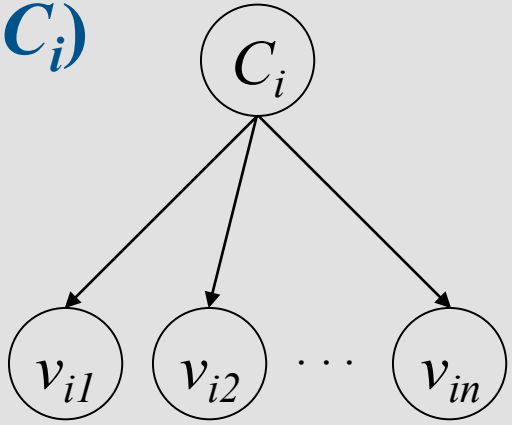
- **Assumes conditional independence of the attribute values for different classes**

$$\Pr(C_i = c \mid v_{i1}, \ldots, v_{in}) = \alpha \prod_{k=1}^{n} \Pr(v_{ik} \mid C_i = c)\Pr(C_i = c)$$

  - where α is a normalizing constant

Naïve Bayes Classifier (II)

- **The parameters: $\Pr(C_i)$ and $\Pr(V_i \mid C_i)$**



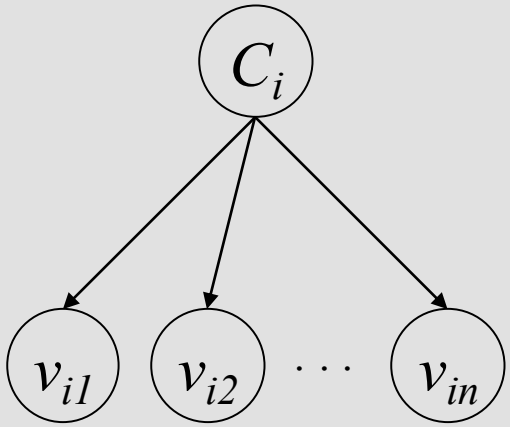
- **BIG questions:**
  - **Prediction/classification: Given that we know the parameters, how do we predict or classify an instance?**
  - **Learning: How do we learn the parameters?**

Naïve Bayes Classifier – Usage

- **The predicted class for a new $x_i = (v_{i1},\ldots,v_{in})$ is:**

$$c^* = \arg\max_c \Pr(c \mid v_{i1},\ldots,v_{in})$$

$$= \arg\max_c \prod_{k=1}^{n} \Pr(v_{ik} \mid c)\Pr(c)$$



Naïve Bayes Classifier – Example

- **4 attributes – outlook, temperature, humidity, wind**
- **2 target classes – YES/NO (play ball)**
- **Probability of a class:**

$c =$	YES	NO
$\Pr(C_i=c)$	9/14	5/14

obtained from the data

- **Calculating $Pr(C_i=YES|v_{i1}=sunny,v_{i2}=hot,v_{i3}=high,v_{i4}=weak)$**

$$\Pr(C_i = YES \mid v_{i1} = sunny, v_{i2} = hot, v_{i3} = high, v_{i4} = weak)$$
$$= \alpha \Pr(v_{i1} = sunny \mid C_i = YES) \times \Pr(v_{i2} = hot \mid C_i = YES) \times$$
$$\Pr(v_{i3} = high \mid C_i = YES) \times \Pr(v_{i4} = weak \mid C_i = YES) \times \Pr(C_i = YES)$$
$$= \alpha \frac{2}{9} \times \frac{2}{9} \times \frac{3}{9} \times \frac{6}{9} \times \frac{9}{14} = \alpha \times 0.007$$

- **This calculation is repeated for all values of c**

Learning the Parameters

Estimating the CPTs $\Pr(c)$ and $\Pr(v \mid c)$

- ***Empirically:*** **use training data**
  - Estimate the probability of a class c
  - For each attribute value w_a, estimate the probability of this value from all instances in a class
- ***Elicitation:*** **ask a person**
  - Usually need domain experts, and sophisticated ways of eliciting probabilities
  - Trouble calibrating

Estimating Parameters Empirically

- **Maximum Likelihood Estimator (MLE):**

$$\Pr_{ML}(var = w_a) = \frac{count(var = w_a)}{\#\ of\ samples} = \frac{|\ var = w_a\ |}{\sum_{i=1}^{m} |\ var = w_i\ |}$$

count

  - where m is the number of values for var

- **Example:** T H T T

  Assume the classes of interest are {H,T}

  - MLE over all samples
    $\Pr_{ML}(H) = {}^{1}/_{4}$, $\Pr_{ML}(T) = {}^{3}/_{4}$
  - MLE of an attribute over one class
    $\Pr_{ML}(green|T) = {}^{2}/_{3}$, $\Pr_{ML}(yellow|T) = {}^{1}/_{3}$, $\Pr_{ML}(red|T) = 0$

Maximum Likelihood Principle

- **Suppose you have data D, and a probabilistic model parameterized by Θ**
  - Need to learn parameter Θ from data D
- ***Likelihood*: The probability of data given the model**
- ***Maximum likelihood*: Choose Θ* which maximizes (the log of) the likelihood function:**

$$\Theta^* := \text{argmax}_\Theta \log \Pr_\Theta(D)$$

Likelihood

  - Set the derivative to 0

$$\frac{\partial}{\partial\theta}[\log \Pr_\theta(D) = 0] \rightarrow \frac{count(var = w_a)}{count\ all\ instances\ of\ var}$$

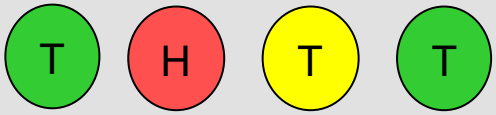
The Sparse Data Problem – Smoothing

- **Not all instances are found in the data set or in a particular class**
  - If value w_a is not found in class c, $w_a=0$ → MLE for $\Pr(w_a|c)=0$
  - If class c or variable var are not found in the training set, MLE for $\Pr(w_a)$ is undefined (denominator is zero)
- ***Expected Likelihood Estimator (ELE)***

$$\Pr_{EL}(var = w_a) \cong \frac{|w_a|+\varepsilon}{\sum_{i=1}^{m}\{|w_i|+\varepsilon\}} = \frac{|w_a|+\varepsilon}{\sum_{i=1}^{m}|w_i|+m\varepsilon}$$

  where m is the number of values for var
  - If a variable is not found in the dataset, ELE is $1/m$
  - ELE is conservative
- **Use Smoothing when estimating the parameters of Naïve Bayes, i.e., $\Pr(C)$ and $\Pr(v|C)$**

Expected Likelihood Estimation – Example



$$\Pr_{ML}(c = H) \cong \frac{|H|}{\sum_1^2 |C_i|} = \frac{1}{1+3} = \frac{1}{4}$$

$$\Pr_{ML}(C) = \left\langle \underset{\text{H}}{\frac{1}{4}}, \underset{\text{T}}{\frac{3}{4}} \right\rangle$$

$$\Pr_{EL}(c = H) \cong \frac{|H| + \varepsilon}{\sum_1^2 |C_i + \varepsilon|} = \frac{|H| + \varepsilon}{\sum_1^2 |C_i| + 2\varepsilon} = \frac{1+\varepsilon}{\{1+3\} + 2\varepsilon} = \frac{1+\varepsilon}{4+2\varepsilon}$$

for ε=1 $\quad \Pr_{EL}(C) = \left\langle \underset{\text{H}}{\frac{1}{3}}, \underset{\text{T}}{\frac{2}{3}} \right\rangle$

MLE and ELE – Text Example

Consider the following dataset for the word that follows "denied the":

Word	Count
allegations	3
reports	2
claims	1
request	1
TOTAL	7

1. Use Maximum Likelihood Estimation to calculate Pr("reports"|"denied the") and Pr("rumours"|"denied the")
2. Repeat these calculations using Expected Likelihood Estimation with ε=0.5 and ε=0.1

MLE and ELE – Text Example Solution

Maximum Likelihood Estimation and Expected Likelihood Estimation

- **MLE**
  - $\Pr(\text{"reports"} \mid \text{"denied the"}) = \frac{c(\text{"denied the reports"})}{c(\text{"denied the"})} = \frac{2}{7}$
  - $\Pr(\text{"rumours"} \mid \text{"denied the"}) = \frac{c(\text{"denied the rumours"})}{c(\text{"denied the"})} = \frac{0}{7}$
- **ELE**
  - $\Pr(\text{"reports"} \mid \text{"denied the"}) = \frac{c(\text{"denied the reports"})}{\sum_X c[(\text{"denied the" } X)+\varepsilon]} = \frac{2+\varepsilon}{7+5\varepsilon}$
  - $\Pr(\text{"rumours"} \mid \text{"denied the"}) = \frac{c(\text{"denied the rumours"})+\varepsilon}{\sum_X c[(\text{"denied the" } X)+\varepsilon]} = \frac{0+\varepsilon}{7+5\varepsilon}$

MLE and ELE – Text Example Solution

Maximum Likelihood Estimation and Expected Likelihood Estimation with ε=1, ε=0.5 and ε=0.1

ε	0 (MLE)	1	0.5	0.1
Pr("reports" | "denied the")	$\frac{2}{7} = 0.286$	$\frac{3}{12} = 0.25$	$\frac{2.5}{9.5} = 0.263$	$\frac{2.1}{7.5} = 0.28$
Pr("rumours" | "denied the")	$\frac{0}{7} = 0$	$\frac{1}{12} = 0.083$	$\frac{0.5}{9.5} = 0.053$	$\frac{0.1}{7.5} = 0.013$

Enhancements to Naïve Bayes

- **Handle missing attribute values**
  - **Categorical**
    - > ignore missing values
    - > consider a missing value (?) to be an additional value
    - > use the most common value of the attribute
    - > probabilistic selection according to the distribution of values
  - **Continuous**
    - > use the mean of the attribute values
- **Handle continuous-valued attributes**

Continuous-valued Attributes

1. **Fit a distribution to the values for attribute A**
   - commonly a Normal distribution
2. **Calculate the probability of the value t of attribute A for item i given the value of class C_j**
   - use a Normal density function

$$\Pr(v_{iA} = \mathrm{t} \mid C_j = \mathrm{c}) \cong \Pr(\mathrm{t} - \frac{\varepsilon}{2} \leq v_{iA} \leq \mathrm{t} + \frac{\varepsilon}{2})$$

$$\cong \varepsilon \frac{1}{\sqrt{2\pi}\,\sigma} e^{-\frac{(t-\mu)^2}{2\sigma^2}}$$

Continuous-valued Attributes – Example

- **For class C_j=*YES*, v_{i2} (temperature) has a Normal distribution with mean μ=*73* and standard deviation σ=*5.23***
- **The probability of temperature 85 F given class *YES* is**

$$\Pr(v_{i2} = 85 | C_j = YES) = \varepsilon \frac{1}{\sqrt{2\pi} \cdot 5.23} e^{-\frac{(85-73)^2}{2 \cdot 5.23^2}}$$

$$= 0.0055\varepsilon$$

Decision Trees versus Naïve Bayes

	Decision Trees (Discriminative)	Naïve Bayes (Generative)
Accuracy	depends on the features of the data	
Robustness	good	good
Generality	YES	assumes independent attributes
Model construction speed (scalability)	slower	faster
Model size	bigger	smaller
Interpretability	higher	lower

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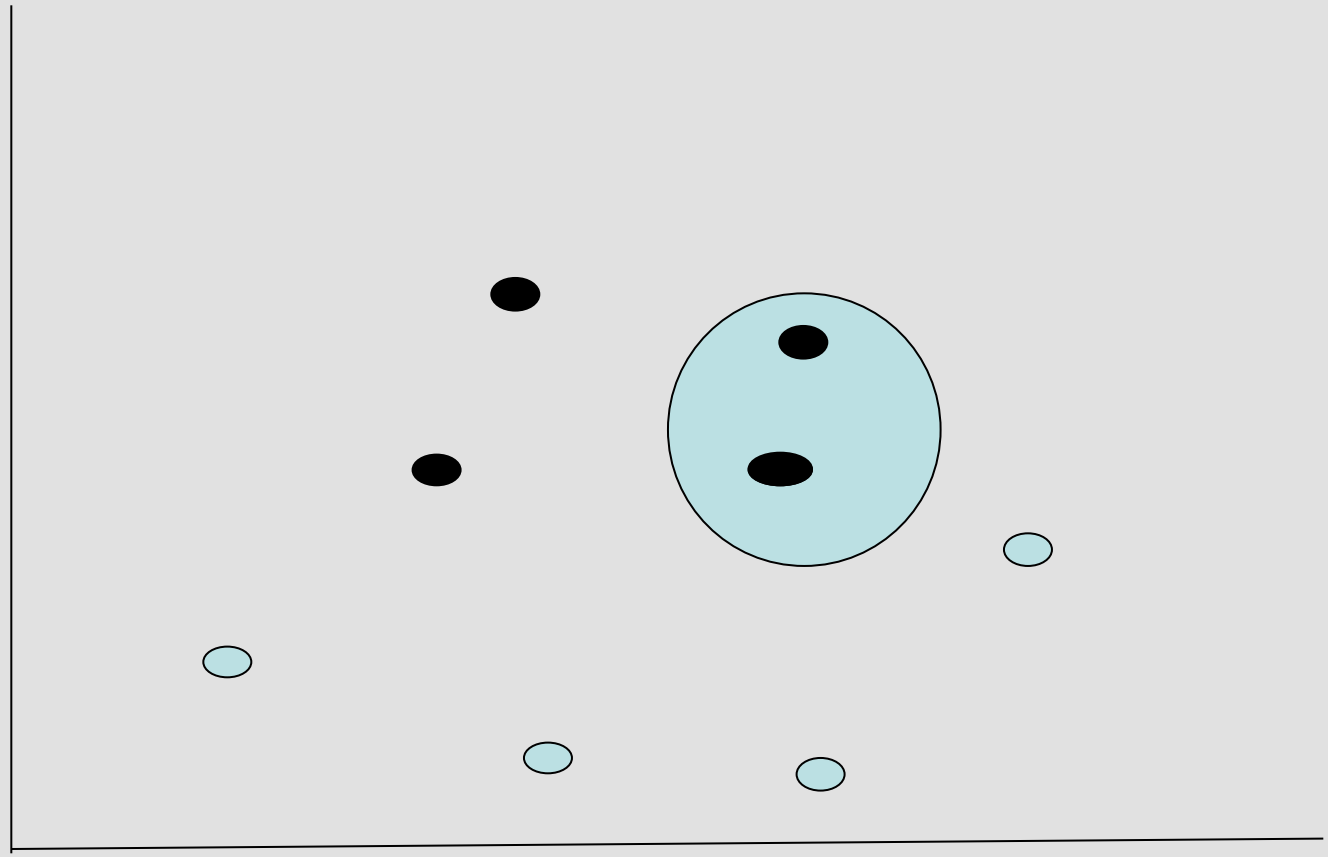
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K Nearest Neighbour (k-NN)

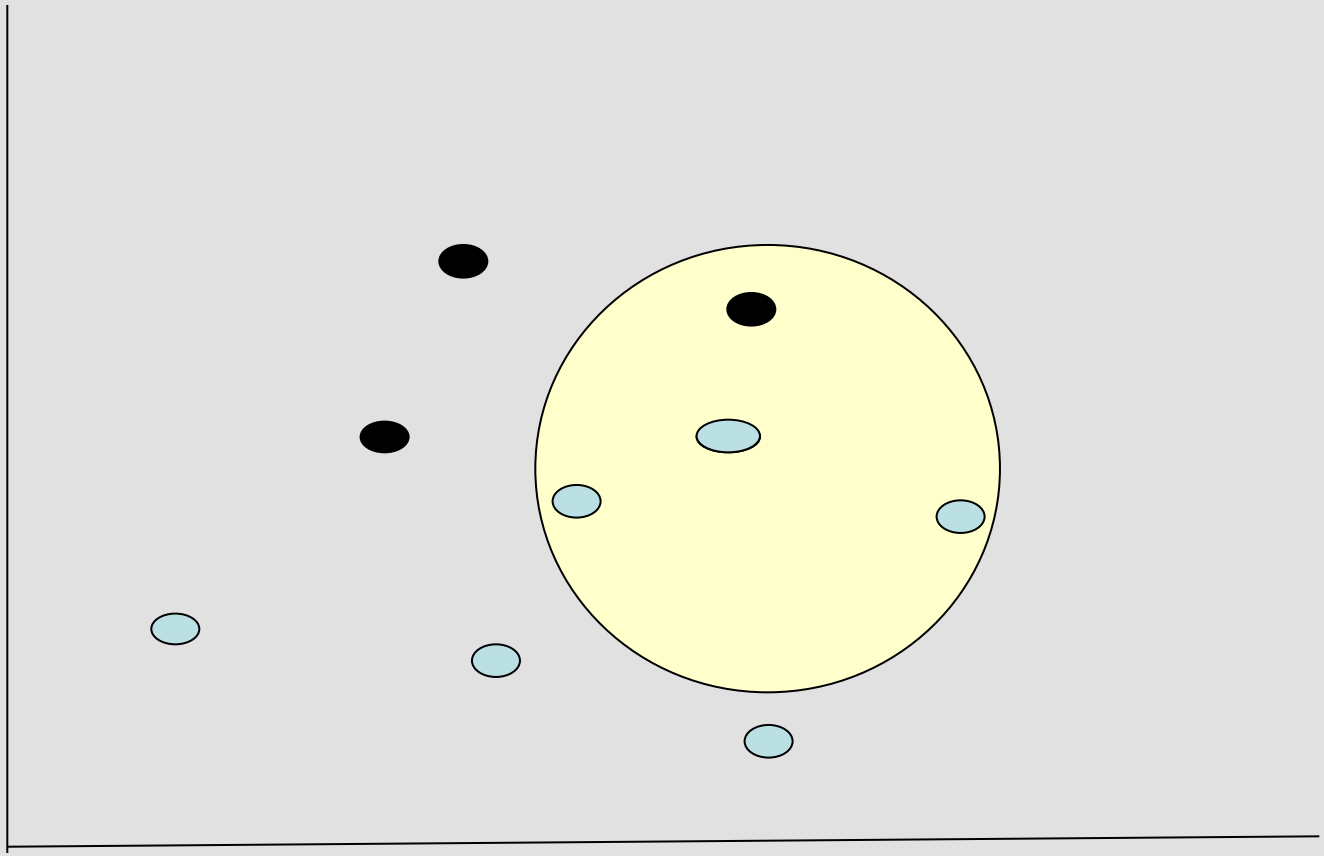
k-Nearest Neighbour (I)

- **All instances correspond to points in an n-dimensional space**
- **Classification is performed**
  - when a new instance arrives
  - by comparing features of the new instance with features of *k* "training" instances that are closest to it in the space (***nearest neighbours***)
- **The target function may be discrete or continuous**
  - **Discrete** – majority vote of the new instance's neighbours
  - **Continuous** – mean value of the k nearest training examples
- **The calculation of the target function may be**
  - Unweighted
  - Weighted (by distance from the new instance)

1-Nearest Neighbour



3-Nearest Neighbour



k-Nearest Neighbour (II)

- **An instance x_i is represented by $(f_{i1}, f_{i2}, ..., f_{in})$**
  - $f_{i,k}$ is the value of the kth feature for x_i
- **Distance measures between two instances x_i and x_j**
  - **Continuous features:** Euclidean distance

$$Ed(x_i, x_j) = \sqrt{\sum_{k=1}^{n} (f_{ik} - f_{jk})^2}$$

  - **Categorical features:** Jaccard coefficient

$$Jc(x_i, x_j) = \frac{|\{f_{i1}, \dots, f_{in}\} \cap \{f_{j1}, \dots, f_{jn}\}|}{|\{f_{i1}, \dots, f_{in}\} \cup \{f_{j1}, \dots, f_{jn}\}|}$$

Must be applied to all features

Computing Similarity – Example

- **Continuous features:**

  x_i={0.7, 30, 80, 10} and x_j={0.2, 32, 85, 40}

  $$Ed(x_i, x_j) = \sqrt{(0.7-0.2)^2 + (30-32)^2 + (80-85)^2 + (10-40)^2}$$

  Smaller is better!

  Need to normalize features

- **Categorical features (Jaccard adaptation):**

  x_i={sunny, hot, high, weak} and x_j={rainy, hot, high, strong}

  $$Jc(x_i, x_j) = \frac{|\{hot, high\}|}{|\{sunny, rainy, hot, high, weak, strong\}|} = \frac{2}{6} = 0.33$$

  Larger is better!

- **Need to combine categorical and continuous features**

Combining Predictions – Same Metric

- **No distance weighting**
- **Distance weighting**
  - Euclidean – weighted average of {Prediction × 1/distance}
  - Jaccard – weighted average of {Prediction × similarity}
- **Example – distance weighting**
  - Euclidean – predictions are 7, 2 and 4, and distances are 1, 2 and 3 respectively.

$$\frac{7\times 1/1 \;+\; 2\times 1/2 \;+\; 4\times 1/3}{1/1 \;+\; 1/2 \;+\; 1/3} = 5.09$$

  - Jaccard – predictions are Y, Y, N, and similarities are 1, 0.6 and 0.6 respectively.

$$Y \times (1 + 0.6) = 1.6Y \qquad 0.6N$$

Combining Predictions – Different Metrics

- **Gower similarity: a metric for computing pairwise similarities between observations when different types occur in the same data set (e.g., nominal, ordinal)**
  - Range: Different 0, Most similar 1
  - **Formulation:**
    - > Numeric element: $f(x_i, yi) = 1 - \frac{|x_i - y_i|}{range}$
    - > Non-numeric element: $f(x_i, yi) = \begin{cases} 1 & if \ \ x_i = y_i \\ 0 & else \end{cases}$
    - > Gower similarity *S*: average of the terms
- **Gower distance:** $\sqrt{1 - S}$

https://statisticaloddsandends.wordpress.com/2021/02/23/what-is-gowers-distance/

Gower Distance – Example

- **Consider two individuals, A and B**

	A	B	Gower Similarity
Age (range: 70)	30	45	$1 - \frac{\lvert 45 - 30 \rvert}{70} = 0.786$
Income (range: 80K)	50K	60K	$1 - \frac{\lvert 60 - 50 \rvert}{80} = 0.875$
Gender	F	M	0

  - Gower similarity: $s = \frac{0.786 + 0.875 + 0}{3} = 0.554$
  - Gower distance: $\sqrt{1 - s} = \sqrt{1 - 0.554} = 0.668$

Parameter Learning

- **Given the "training data" and the distance function, there is no training**
  - The algorithm memorizes the training data
  - As data comes in, the "model size" grows
- **In some cases, the distance function is parameterized and its parameters are learnt**

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Ensemble Learning

Ensemble Learning – Definition

- **Select a collection, or *ensemble*, of hypotheses $h_1, h_2, \ldots, h_n$, and combine their predictions by averaging, voting, or by another level of machine learning**
  - base models – individual hypotheses
  - ensemble model – their combination
- **Why ensembles?**
  - Reduce bias
  - Reduce variance

Creating ensembles

- **Bagging – generate K distinct training sets by sampling with replacement from the original training set**
- **Random forests – vary the attribute choices by computing information gain for only a subset of the attributes at each split**
- **Stacked generalization (stacking) – combine multiple base models from different model classes trained on the same data**
- **Boosting – iteratively train, and increase the weights of mis-classified training samples**

Summary – Supervised Machine Learning

- **Discrete:**
  - K nearest neighbour — non-parametric — non-linear

  Generative
  - Naïve-Bayes — parametric — linear

  Discriminative
  - Decision Trees — non-parametric — non-linear
  - Logistic regression — parametric — linear
  - Perceptron — parametric — linear
  - Multi-layer perceptron — parametric — non-linear
- **Continuous:**
  - Regression
    - > Models based on least square error (parametric)

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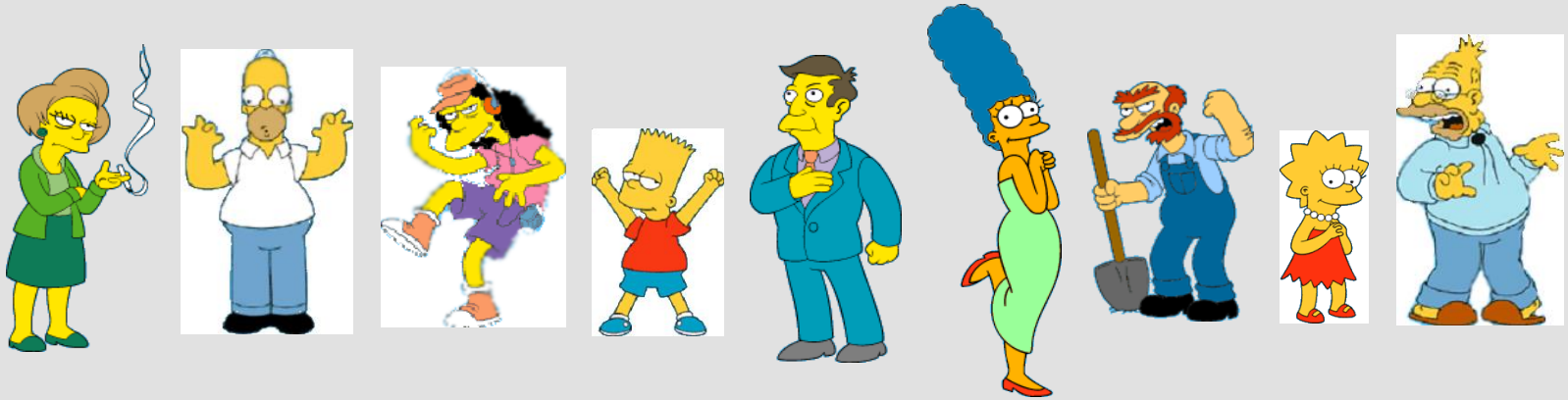
Unsupervised Machine Learning – Clustering

Slides adapted from Eamonn Keogh, Dan Klein and Kevin Korb

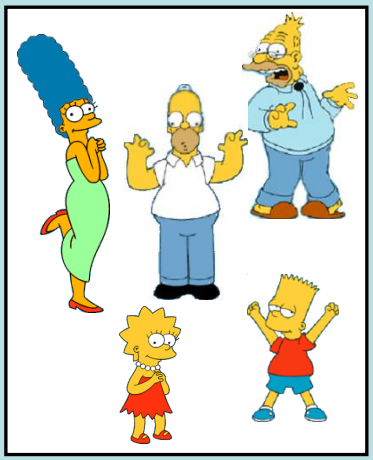
What is Clustering?

- **Organizing data into classes such that there is**
  - high intra-class similarity
  - low inter-class similarity
- **Finding the class labels and the number of classes directly from the data**

What is a Natural Grouping of these Objects?



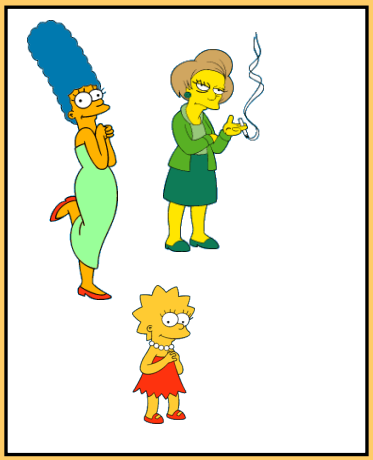
Clustering is subjective



Simpson's Family



School Employees



Females



Males

What is Similarity?

- ***The quality or state of being similar; likeness; resemblance; as, a similarity of features.***

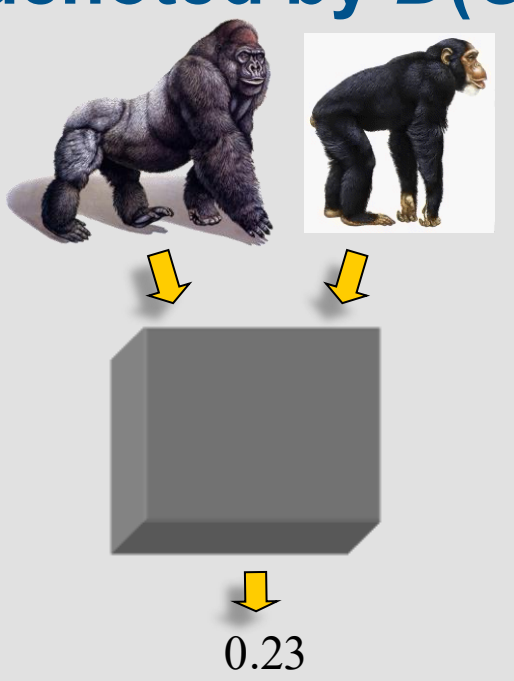
  Webster's dictionary

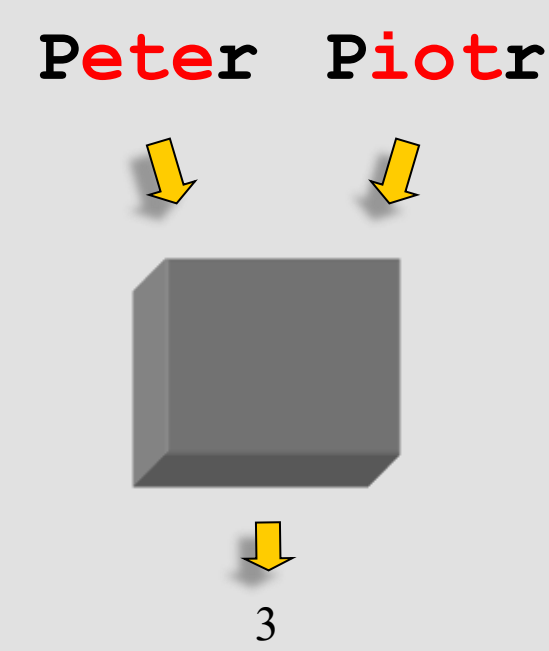
- **Similarity is hard to define, but…**
  "*We know it when we see it*"

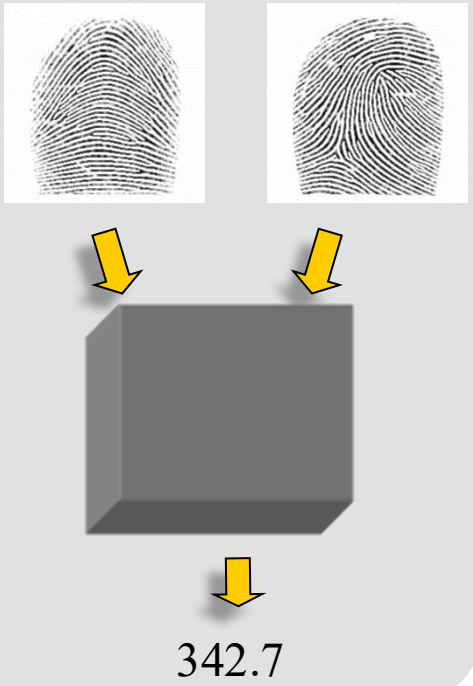


Distance Measures (I)

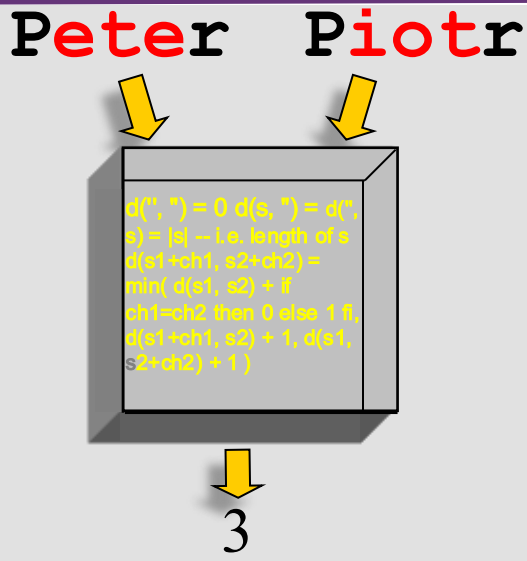
Definition: Let O_1 and O_2 be two objects from the universe of possible objects. The distance (dissimilarity) between O_1 and O_2 is a real number denoted by $D(O_1,O_2)$







Distance Measures (II)



When we peek inside one of these black boxes, we see a function of two variables

What properties should a distance measure have?

- $D(A,B) = D(B,A)$ *Symmetry*
- $D(A,A) = 0$ *Constancy of Self-Similarity*
- $D(A,B) = 0$ iff $A = B$ *Positivity (Separation)*
- $D(A,B) \leq D(A,C) + D(B,C)$ *Triangle Inequality*

Measuring Similarity

Edit distance: Transform one of the objects into the other, and measure how much effort it took

Distance between Patty and Selma:

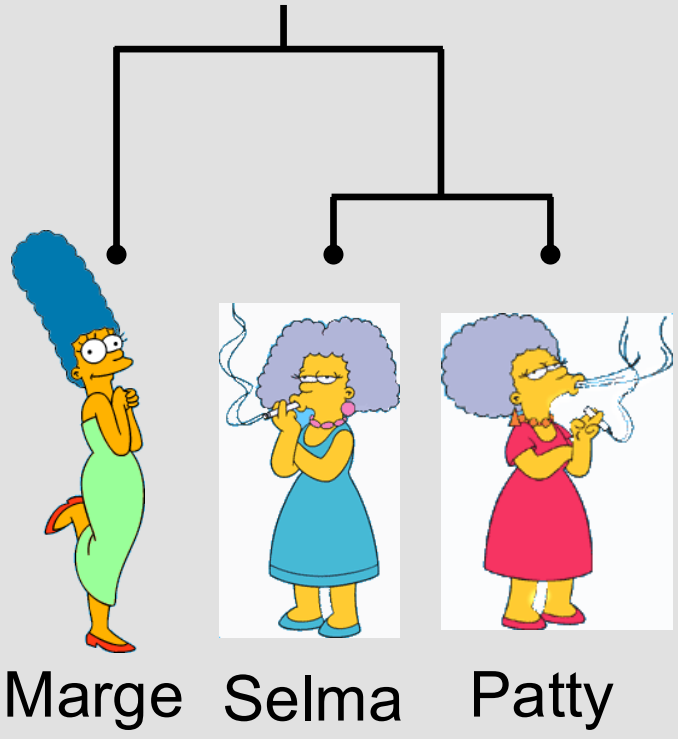
Change dress color	1 point
Change earring shape	1 point
Change hair part	1 point

D(Patty,Selma) = 3

Distance between Marge and Selma:

Change dress color	1 point
Add earrings	1 point
Decrease height	1 point
Take up smoking	1 point
Gain weight	1 point

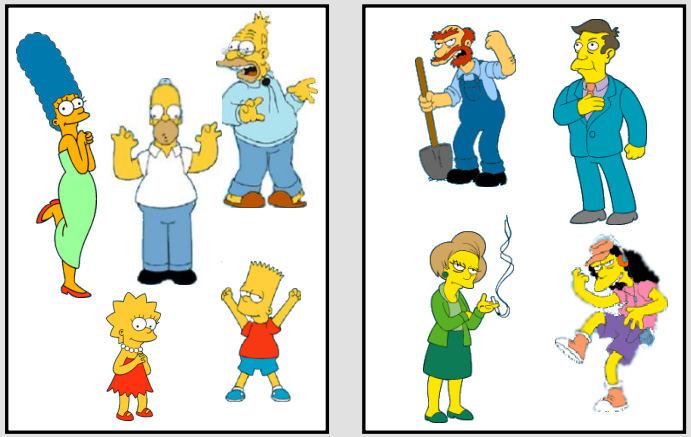
D(Marge,Selma) = 5



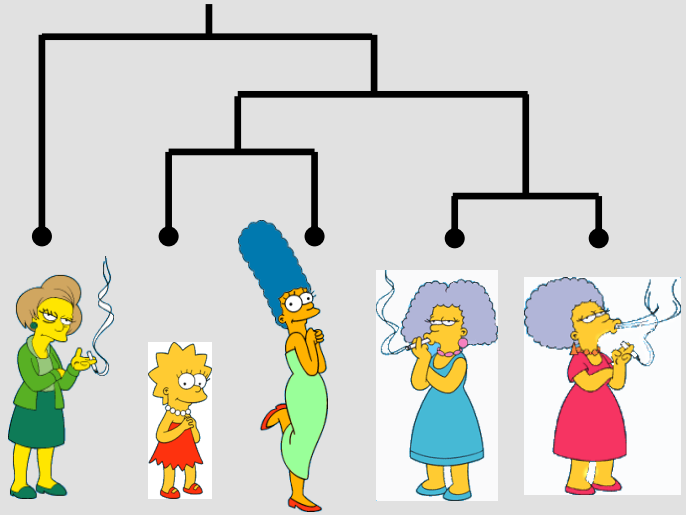
Two Types of Clustering

- **Partitional: Construct various partitions and then evaluate them by some criterion**
- **Hierarchical: Create a hierarchical decomposition of the set of objects using some criterion**

Partitional



Hierarchical



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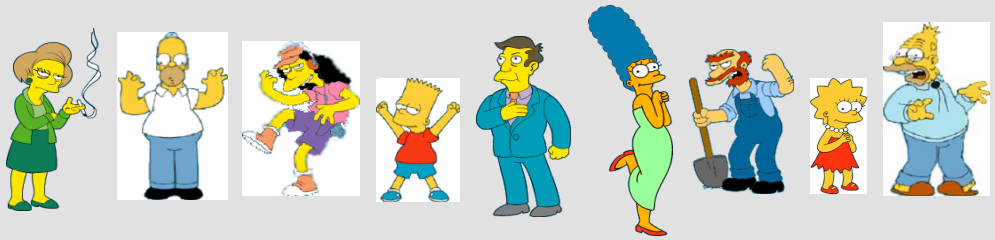


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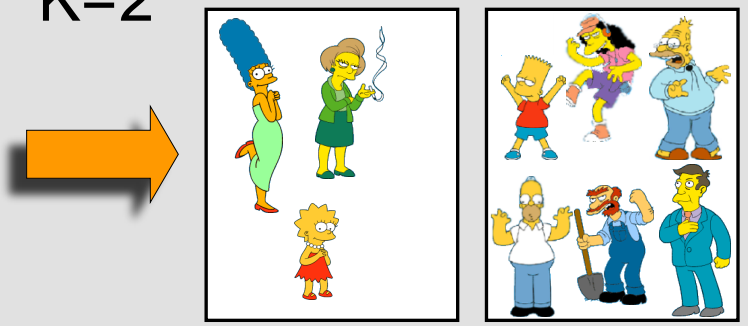
K-means Algorithm

Partitional Clustering

- **Non-hierarchical, each instance is placed in exactly one of K non-overlapping clusters**
- **Produces only one set of clusters**
  ➔ the user normally has to input the desired number of clusters K
- **A goodness score is used to select the best configuration**



K=2



Squared Error

Squared error for cluster K_i

of items in cluster K_i

Data point j in cluster K_i

$$se_{K_i} = \sum_{j=1}^{m_i} ||x_{ij} - C_i||^2$$

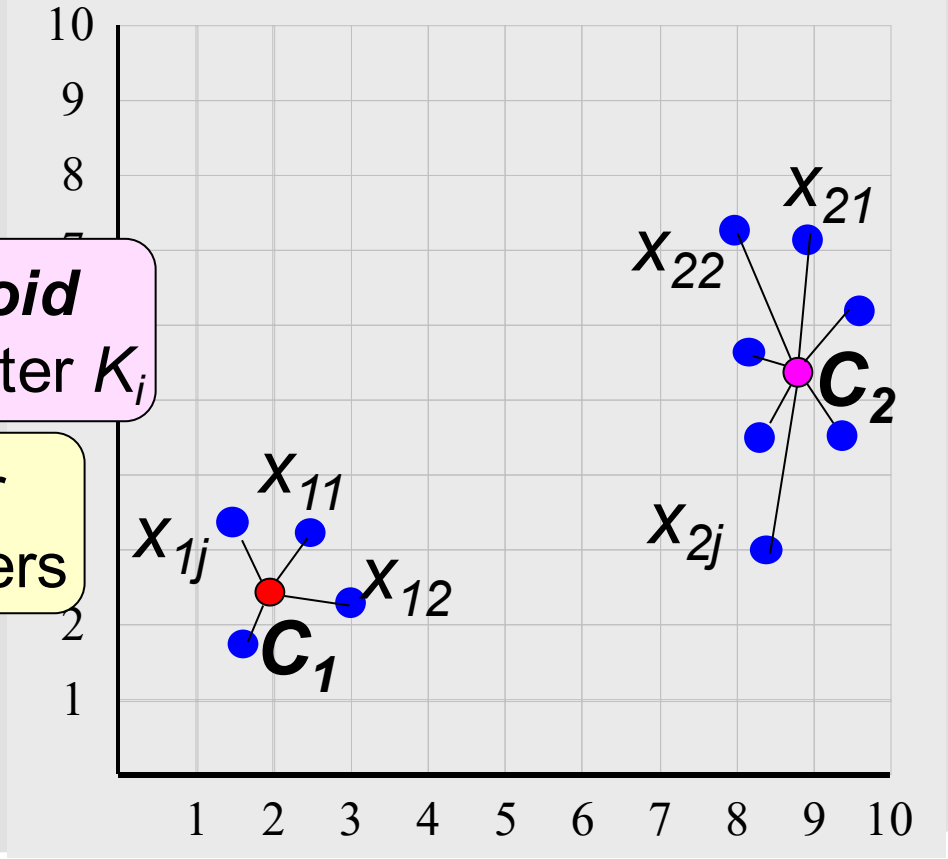
Centroid of cluster K_i

$$se_K = \sum_{i=1}^{K} se_{K_i}$$

Number of clusters

Objective Function:

min se_K

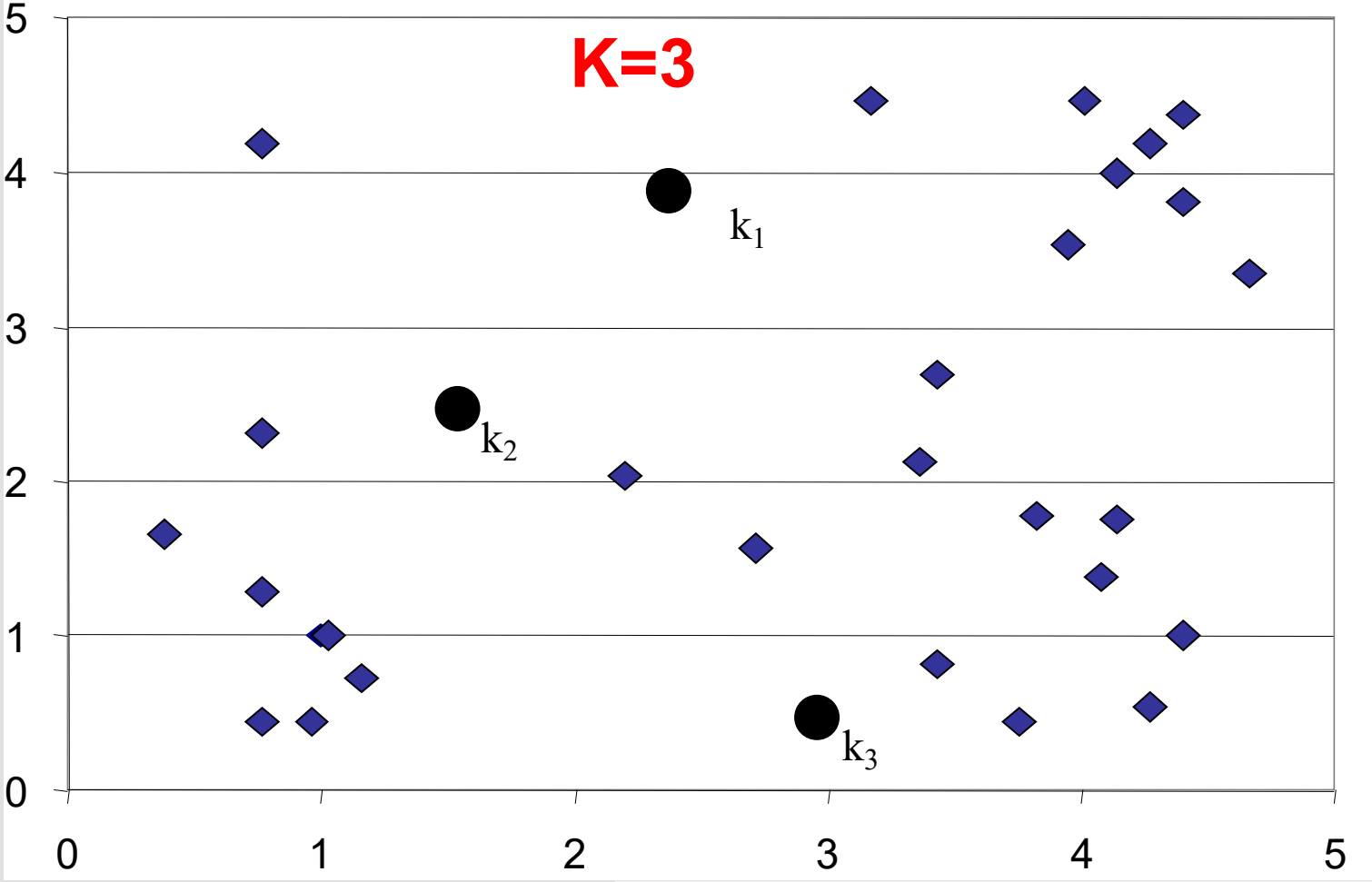


K-means Algorithm

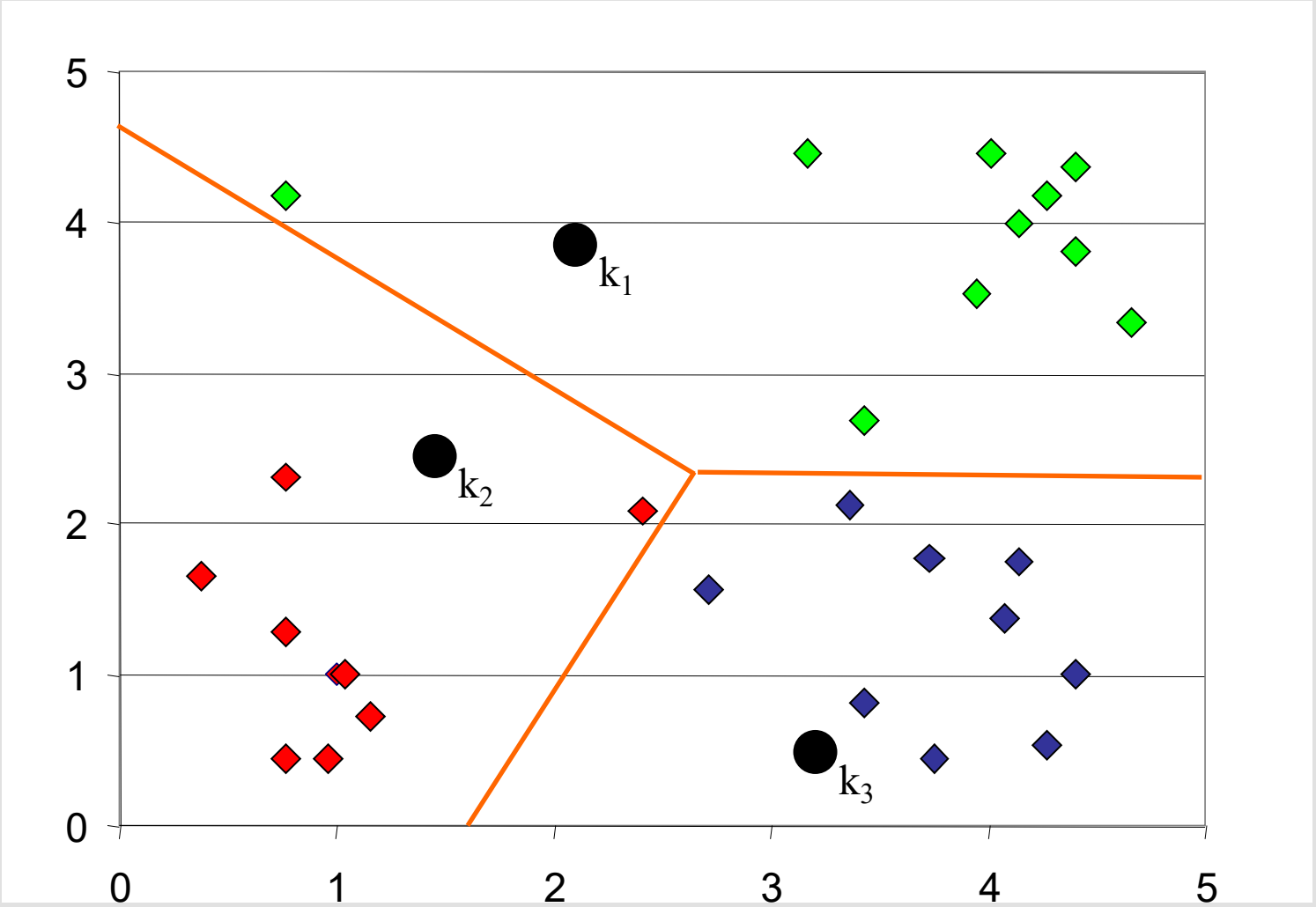
1. Decide on a value for *k*
2. Initialize the *k* cluster *centroids* (randomly, if necessary)
3. Decide the class memberships of the *N* objects by assigning them to the nearest cluster centroid
4. If none of the *N* objects changed membership in the last iteration, exit
5. (Otherwise) Re-estimate the *k* cluster centroids, assuming the memberships found above are correct
6. Goto Step 3

K-means Clustering (1): Steps 1 and 2

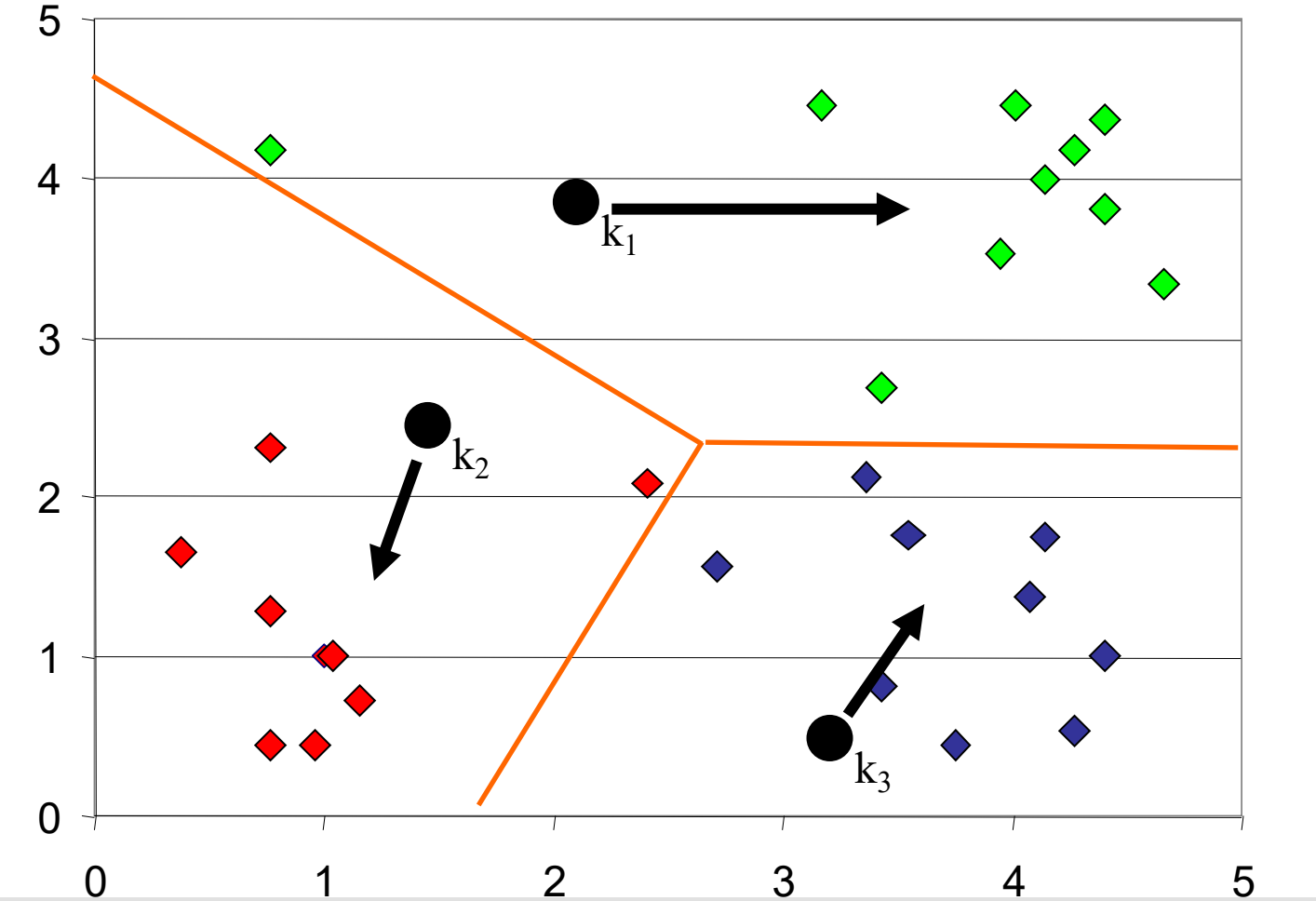
Distance Metric: Euclidean Distance



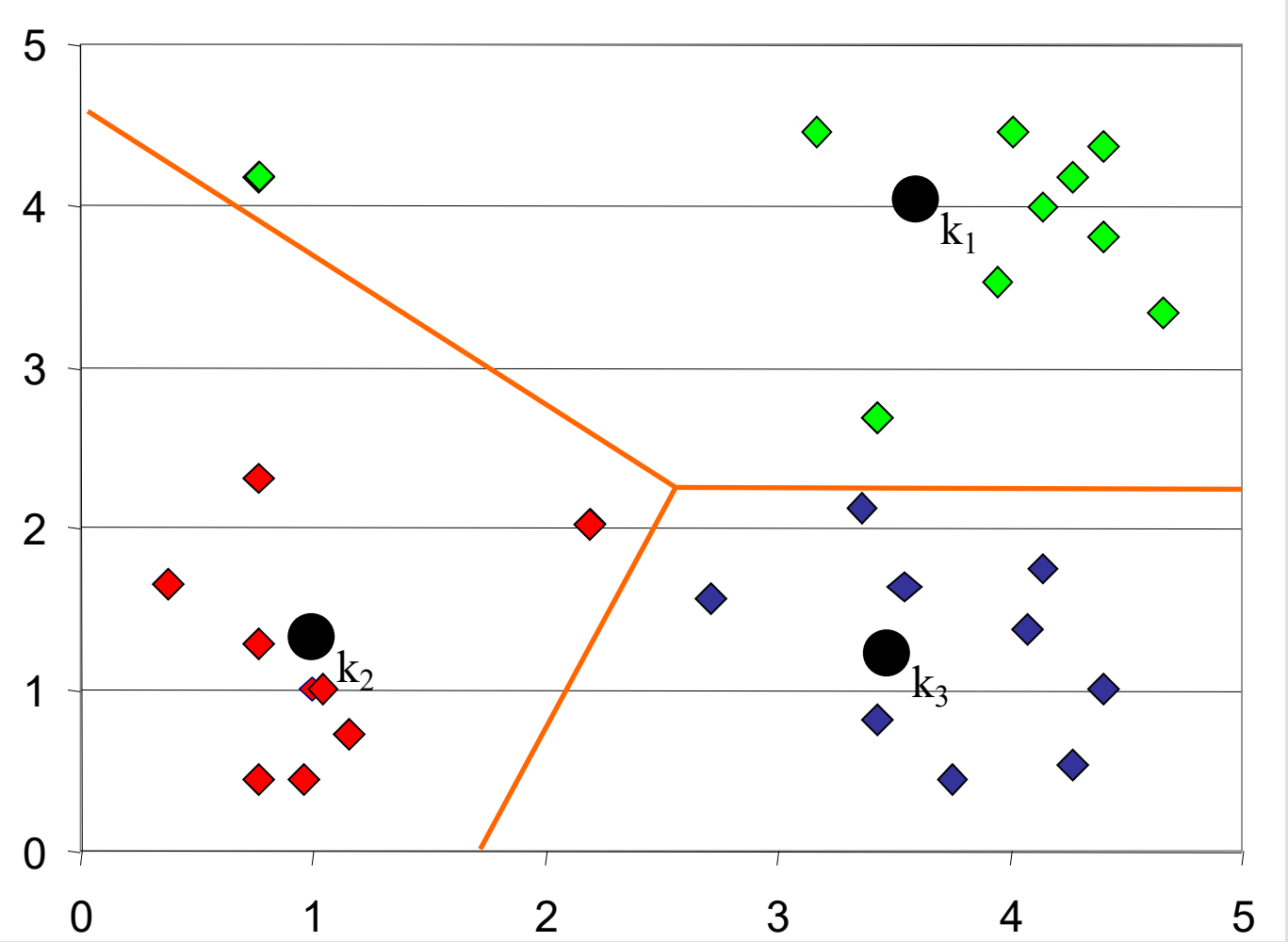
K-means Clustering (1): Step 3



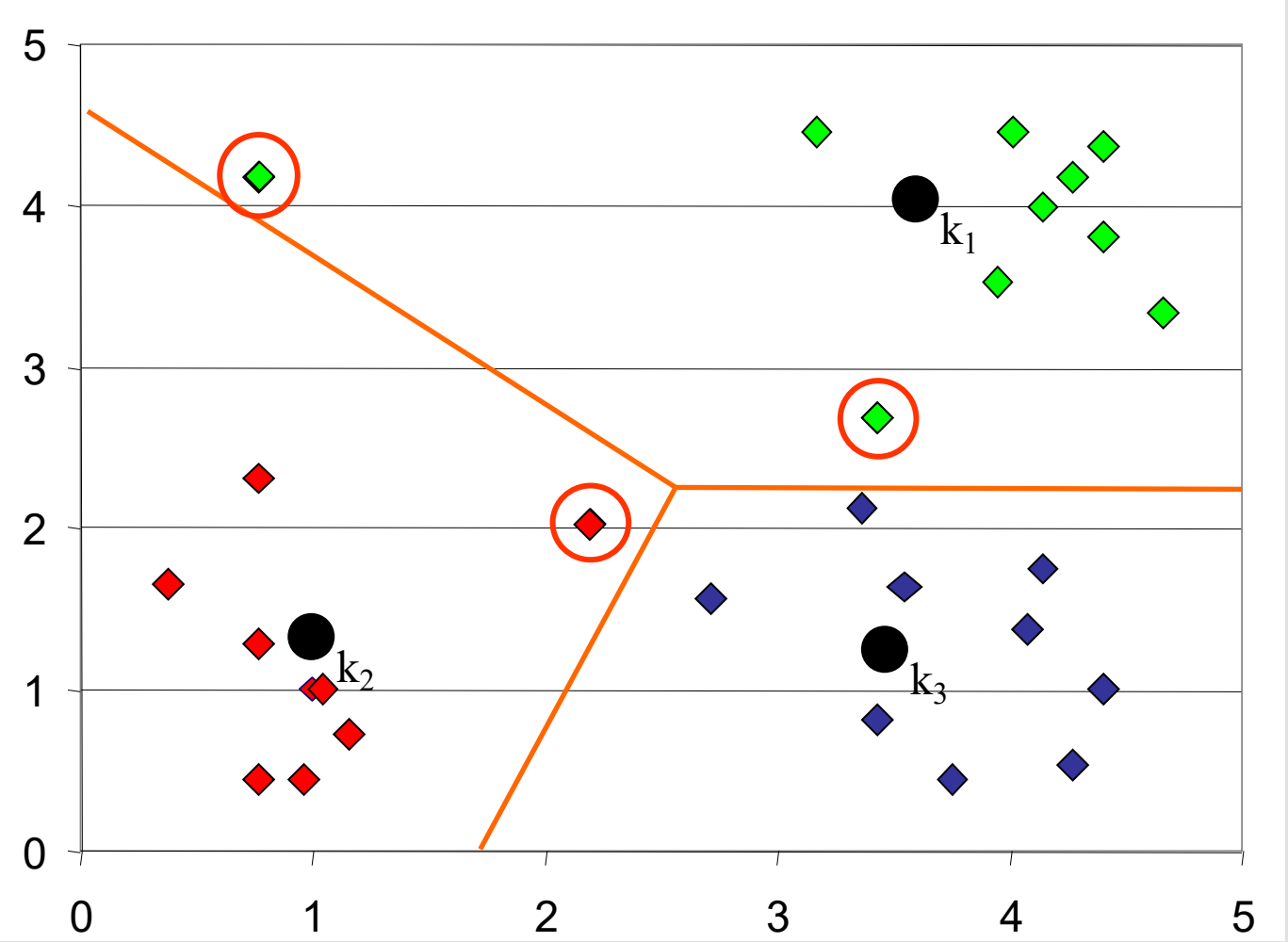
K-means Clustering (1): Step 5



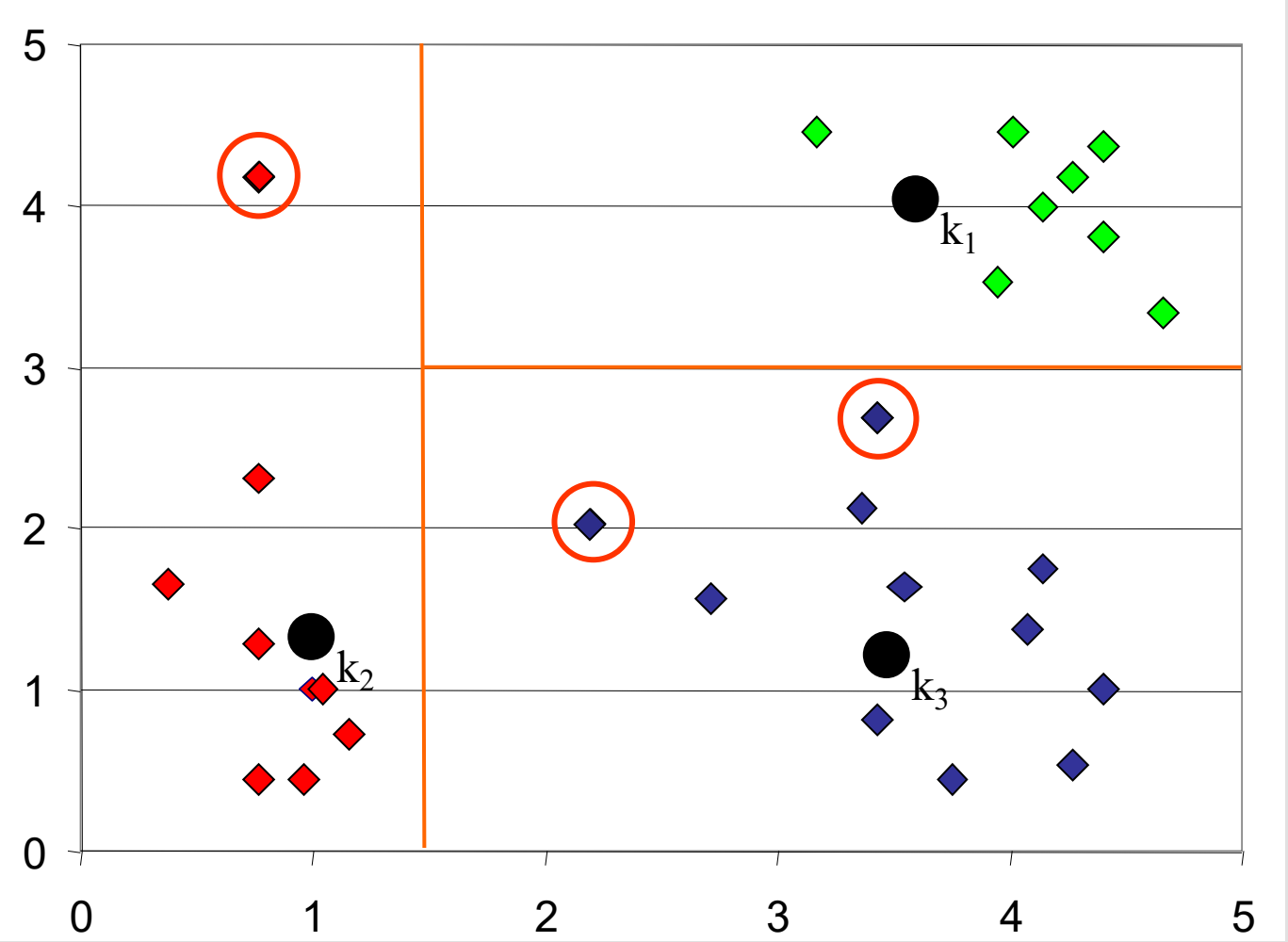
K-means Clustering (1): Step 5



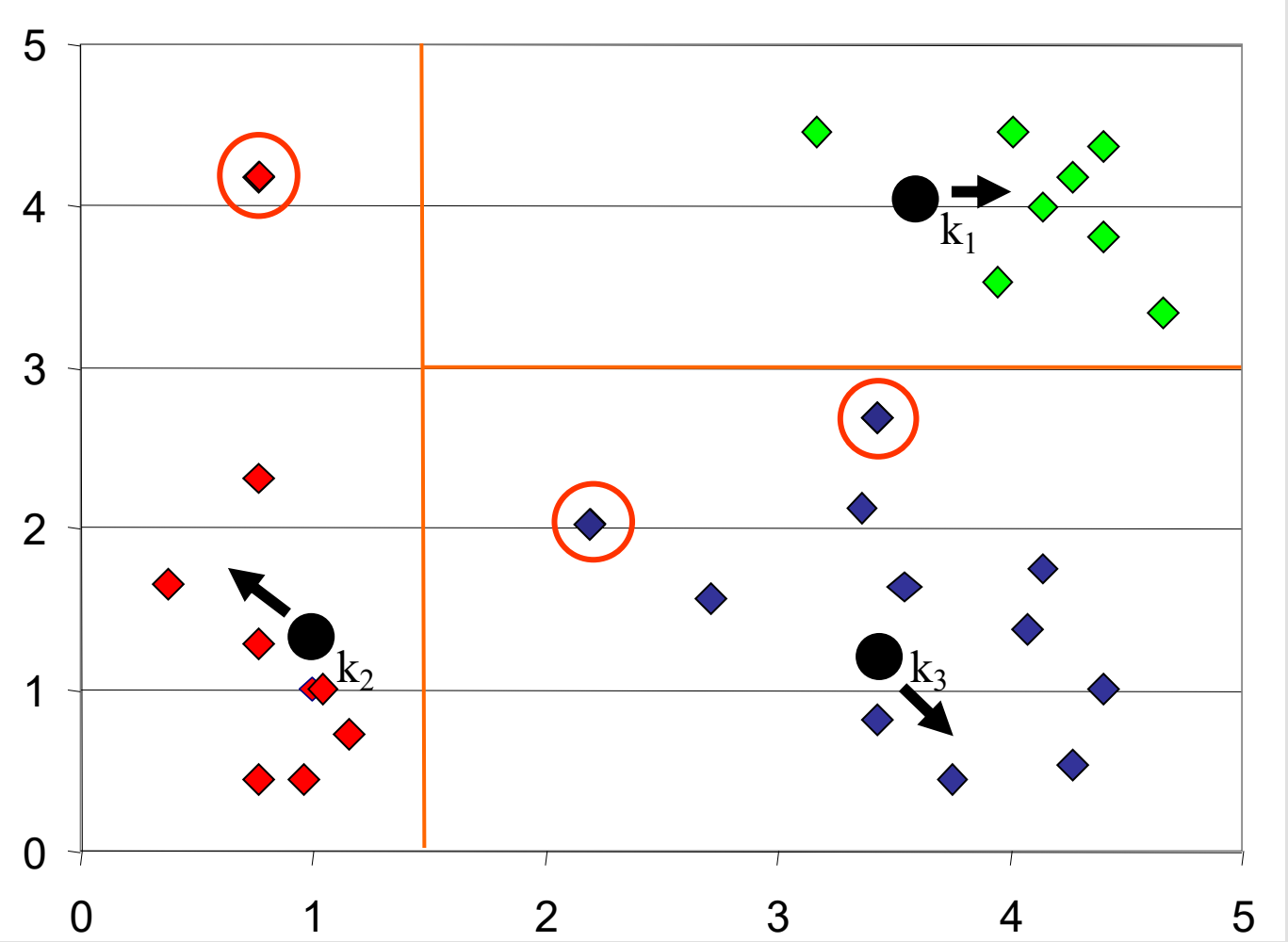
K-means Clustering (2): Step 3



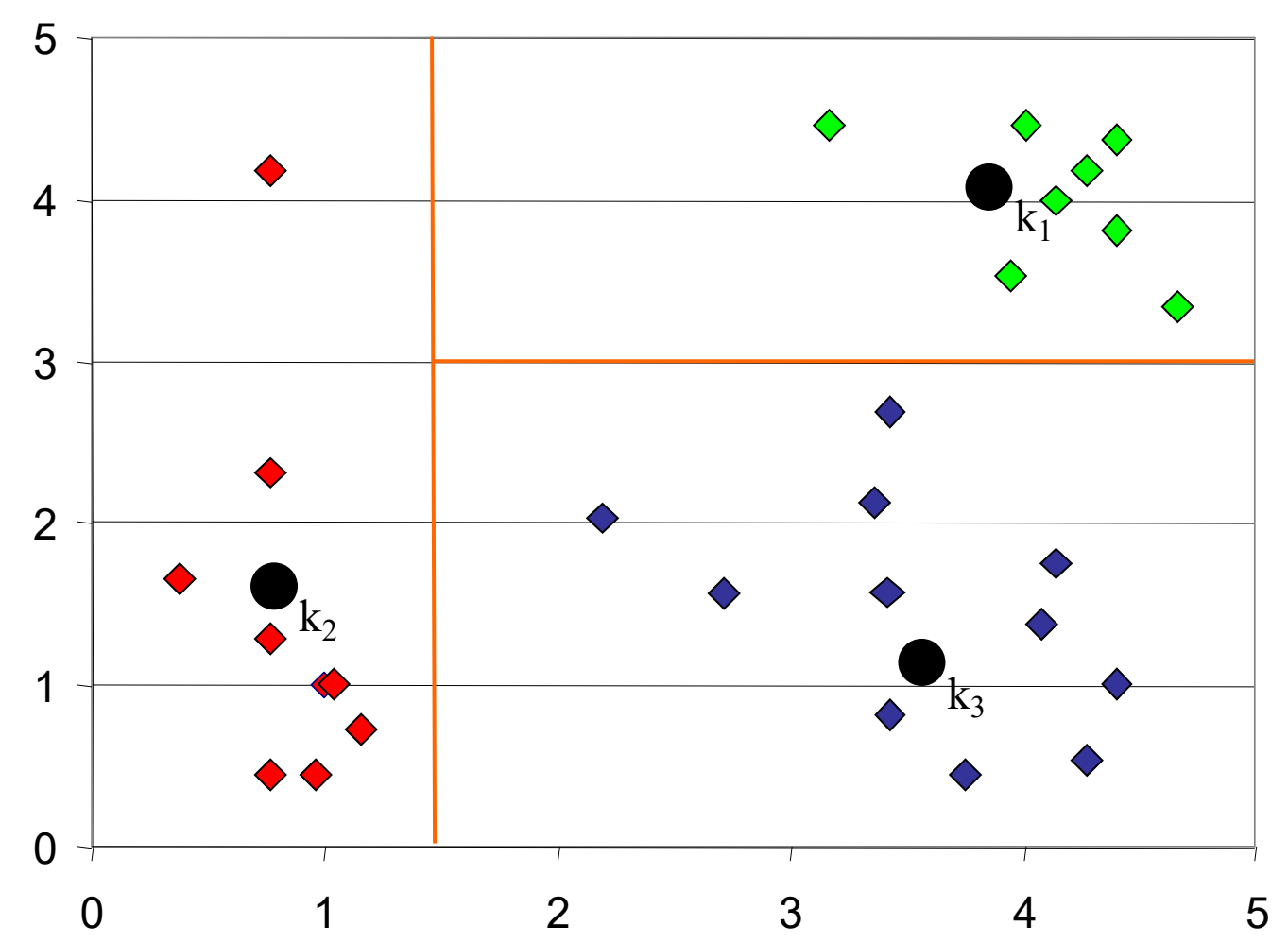
K-means Clustering (2): Step 3



K-means Clustering (2): Step 5

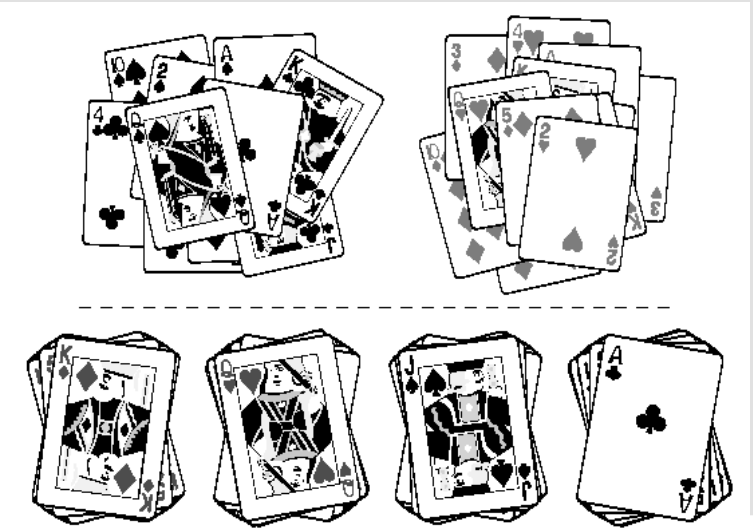


K-means Clustering (2): Step 5



The Meaning of K

- **Clusters describe underlying structure in data**
  - but structures in data can exist at different levels
- **In many cases, there is no a priori reason to select a particular value for K**
  - should consider several K-s, but different values of K can lead to different clusterings



K = 2

K = 4

Normalization

- **Problem: distance will be dominated by attributes with a large magnitude**
- **Example: in which cluster do we put age=25 & income=$30,000?**
  - Centroid 1: age=26 & income=$25,000
  - Centroid 2: age=80 & income=$34,500
- **Solution: normalize the features**

Selecting the best configuration

- **Silhouette score measures an item's intra-cluster similarity and inter-cluster dissimilarity**
- **Range is [-1,+1]**
  - High value: good intra-cluster match, poor inter-cluster match
- **Item $i \in C_k$**
  - Intra-cluster similarity of i $\quad a(i) = \frac{1}{|C_k|-1} \sum_{j \in C_k, i \neq j} d(i,j)$
  - Inter-cluster similarity of i $\quad b(i) = \min_{C_l \neq C_k} \frac{1}{|C_l|} \sum_{j \in C_l} d(i,j)$

$$s(i) = \begin{cases} \dfrac{b(i) - a(i)}{\max\{a(i), b(i)\}} & \text{if } |C_k| > 1 \\ 0 & \text{if } |C_k| = 1 \end{cases}$$

- **Compute the average silhouette score for all items**

Summary of the K-Means Algorithm

- **Advantages**
  - ***Relatively efficient***: $O(tKn)$, where n is # objects, K is # clusters, and t is # iterations. Normally, K, $t << n$
  - ***Global optimum*** may be found using techniques such as *deterministic annealing* and *genetic algorithms*
- **Disadvantages**
  - Need to specify K in advance
  - Applicable only when *mean* is defined
    - > Extensions for categorical data: K-Mode and K-Prototype
  - Does not deal well with overlapping clusters
  - Unable to handle noisy data and outliers
    - > Outliers can pull cluster centers

Reading

- **Supervised learning –** Russell, S. and Norvig, P. (2022), *Artificial Intelligence – A Modern Approach* (4th ed), Prentice Hall
  - Naïve Bayes – Chapter 12.6, 21.2.1-21.2.2
  - k Nearest Neighbours – Chapter 19.7.1
  - Ensemble learning – Chapter 19.8.1-19.8.4
- **Unsupervised learning and K-means clustering –**
  - http://en.wikipedia.org/wiki/K-means_clustering
  - https://sites.google.com/site/dataclusteringalgorithms/k-means-clustering-algorithm